

Lecture: 11 Successive Differentiation

By Dr. A.K. Misra

1. Definition

Let y be a function of x , its differential coefficient $\frac{dy}{dx}$ will be in general a function of x which can be differentiated. The differential coefficient of $\frac{dy}{dx}$ is called the second differential coefficient of y . The differential coefficient of the second differential coefficient is called the third differential coefficient, and so on. The successive differential coefficients of y are denoted by

$$\frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \dots,$$

and the n th differential coefficient of y being $\frac{d^ny}{dx^n}$.

- If $y = (ax + b)^m$, then

$$y_1 = m.a(ax + b)^{m-1}, y_2 = m(m-1)a^2(ax + b)^{m-2}, \text{etc.}$$

Hence,

$$D^n(ax + b)^m = m(m-1)(m-2)\dots(m-n+1)a^n(ax + b)^{m-n}.$$

In particular,

$$D^n(ax + b)^{-1} = (-1)^n n! a^n (ax + b)^{-n-1}.$$

If m is a positive integer, the $(m+1)$ th and all the differential coefficients of $(ax + b)^m$ would be zero.

- If $y = e^{ax}$, then

$$y_1 = ae^{ax}, y_2 = a^2e^{ax}, \text{etc.}$$

In general,

$$D^n(e^{ax}) = a^n e^{ax}.$$

- If $y = a^x$, then

$$y_1 = (\log a)a^x, y_2 = (\log a)^2 a^x, \text{etc.}$$

In general,

$$D^n a^x = (\log a)^n a^x.$$

- If $y = \log(ax + b)$, then

$$y_1 = \frac{a}{(ax + b)}, y_2 = \frac{-a^2}{(ax + b)^2}, y_3 = \frac{(-1)(-2)a^3}{(ax + b)^{-3}}, \text{etc.}$$

In general,

$$D^n \log(ax + b) = \frac{(-1)^n (n-1)! a^n}{(ax + b)^n}.$$

- If $y = \sin(ax + b)$,

$$y_1 = a \cos(ax + b) = a \sin\left(ax + b + \frac{\pi}{2}\right)$$

,

$$y_2 = a^2 \cos\left(ax + b + \frac{\pi}{2}\right) = a^2 \sin(ax + b + \pi),$$

$$y_3 = a^3 \sin\left(ax + b + \frac{3\pi}{2}\right), \text{etc.}$$

In general,

$$D^n \sin(ax + b) = a^n \sin\left(ax + b + \frac{n\pi}{2}\right).$$

- If $y = \cos(ax + b)$ then,

$$y_1 = -a \sin(ax + b) = a \cos\left(ax + b + \frac{\pi}{2}\right),$$

$$y_2 = -a^2 \sin\left(ax + b + \frac{\pi}{2}\right) = a^2 \cos(ax + b + \pi)$$

$$y_3 = a^3 \cos\left(ax + b + \frac{3\pi}{2}\right), \text{etc.}$$

In general,

$$D^n \cos(ax + b) = a^n \cos\left(ax + b + \frac{n\pi}{2}\right).$$

If we put $a = 1$ and $b = 0$ in above we have,

$$D^n \sin x = \sin\left(x + \frac{n\pi}{2}\right),$$

$$D^n \cos x = \cos\left(x + \frac{n\pi}{2}\right).$$

- If $y = e^{ax} \sin(bx + c)$, then

$$y_1 = e^{ax} b \cos(bx + c) + ae^{ax} \sin(bx + c).$$

Putting $a = r \cos \phi$ and $b = r \sin \phi$, we have

$$y_1 = re^{ax} \sin(bx + c + \phi).$$

Similarly,

$$y_2 = r^2 e^{ax} \sin(bx + c + 2\phi), \text{ etc.}$$

In general,

$$D^n \{e^{ax} \sin(bx + c)\} = r^n e^{ax} \sin(bx + c + n\phi),$$

where $r = (a^2 + b^2)^{1/2}$, and $\phi = \tan^{-1}(b/a)$.

Similarly,

$$D^n \{e^{ax} \cos(bx + c)\} = r^n e^{ax} \cos(bx + c + n\phi),$$

where $r = (a^2 + b^2)^{1/2}$, and $\phi = \tan^{-1}(b/a)$.

For finding the n th differential coefficient of a fraction whose numerator and denominator are both rational algebraic functions, the given fraction must be decomposed into its partial fractions.

Example. Find the n th differential coefficient of

$$\frac{x^4}{(x-1)(x-2)}.$$

Solution.

$$\frac{x^4}{(x-1)(x-2)} = x^2 + 3x + 7 + \frac{15x - 14}{(x-1)(x-2)},$$

$$\frac{x^4}{(x-1)(x-2)} = x^2 + 3x + 7 - \frac{1}{(x-1)} + \frac{16}{(x-2)}.$$

The n th derivative of $x^2 + 3x + 7$ will be zero if $n > 2$, therefore if $n > 2$,

$$\frac{d^n}{dx^n} \frac{x^4}{(x-1)(x-2)} = -(-1)^n \cdot n! (x-1)^{-n-1} + (-1)^n \cdot n! (x-2)^{-n-1}.$$

The n th differential coefficient of $\frac{x^4}{(x-1)(x-2)}$ is

$$(-1)^n (n!) \{16(x-2)^{-n-1} - (x-1)^{-n-1}\}.$$

Example. If $y = e^{ax} \sin bx$, prove that

$$y_2 - 2ay_1 + (a^2 + b^2)y = 0.$$

Solution. Here, we have $y = e^{ax} \sin bx$,

$$\Rightarrow y_1 = ae^{ax} \sin bx + e^{ax} b \cos bx$$

$$\Rightarrow y_1 = ay + be^{ax} \cos bx \Rightarrow y_1 - ay = be^{ax} \cos bx$$

Differentiating this with respect to x , we have

$$y_2 - ay_1 = a \underbrace{be^{ax} \cos bx}_{=y_1 - ay} + be^{ax}(-b \sin bx)$$

$$\Rightarrow y_2 - ay_1 = a(y_1 - ay) - b^2y$$

$$\Rightarrow y_2 - ay_1 = ay_1 - a^2y - b^2y$$

$$\Rightarrow y_2 - 2ay_1 + (a^2 + b^2)y = 0. \text{ Hence proved.}$$

Reference Book: Differential Calculus, Dr. Gorakh Prasad. Pothishala Private LTD.